Handout #5
Thursday, 24 September 2026



Autumn Quarter 2026

Problem Set 1

Due: Thursday, 1 October 2026 at 1:30pm via Gradescope
Topics: Circular Restricted Three Body Problem trajectories, Lagrange points

1.1) (30 points) In honor of the New Horizons spacecraft's flyby of Pluto on 14 July 2015, you will map out some elements of the synodic system comprised of Pluto and its largest moon, Charon. Use μ_{Pluto} = 870.9 [km^3/sec^2] , μ_{Charon} = 101.0 [km^3/sec^2] , and a Pluto-Charon distance of ${}^{Pluto}r^{Charon}$ = 19,571.0 [km] . Mean equatorial radii are r_{Pluto} = 1,184.0 [km] and r_{Charon} = 603.5 [km] . Assume the orbits are circular. ***Do not forget to submit your MATLAB code and plots!***

a) What is the period of Charon's orbit around Pluto?

b) Does the system barycenter lie within Pluto, or between Pluto and Charon?

c) What are the coordinates (in the synodic frame) of L_1? Feel free to assume that L_1 has $y = 0$ and $z = 0$, and to work with scalar quantities.

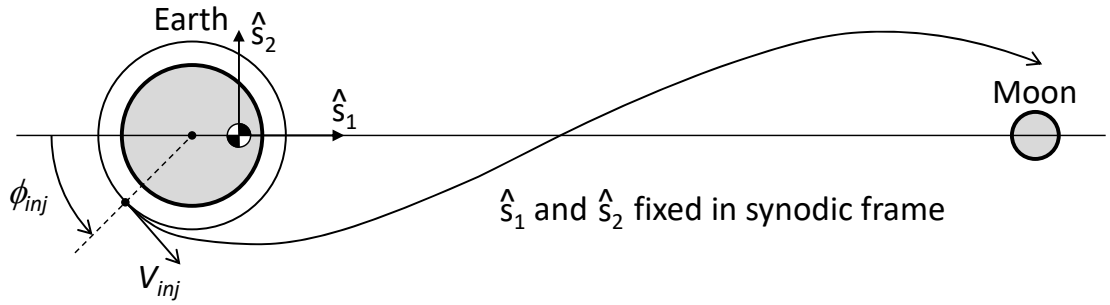
You *may* need to solve for the zero of a fifth order polynomial (with MATLAB's **`fzero`** function for example). Be careful about signs when setting up the quintic equation. For example, $\frac{a}{|a|^3}$ does not always simplify to $\frac{1}{a^2}$; sometimes it simplifies to $-\frac{1}{a^2}$ depending on the sign of a. The expression $\frac{a}{|a|^3} + \frac{b}{|b|^3}$ can have four possible simplifications depending on the signs of a and b.

You will reuse your code for finding L_1 on Problem Set 2.

d) Plot some contours of effective potential energy PE_{eff} (it's also OK to plot contours of the Jacobi constant C with zero velocity; they differ only by a constant). The values corresponding to the contours need not be labeled, but there should be enough contours (at least eight) to get a feel for what the surface looks like. On your plot, show the location of L_1 (and optionally, any other Lagrange points if you like). See MATLAB's **`contour`** function for more information.

1.2) (70 points) Here you'll simulate a "free return" trajectory similar to those used by the early Apollo missions to the Moon. You can assume that the Earth and Moon are perfectly spherical and that together they behave as a circular restricted three body problem.
Use $\mu_{Earth} = 398{,}600$ [km^3/sec^2] , $\mu_{Moon} = 4{,}903$ [km^3/sec^2] , and an Earth-Moon distance of ${}^{Earth}r^{Moon} = 384{,}400$ [km] . Mean equatorial radii are $r_{Earth} = 6{,}378.0$ [km] and $r_{Moon} = 1{,}738.0$ [km] .

Assume the spacecraft begins in a 200 [km] altitude circular LEO parking orbit that is coplanar with the Moon's orbit around Earth. At time $t = 0$, a tangential Trans Lunar Injection (TLI) burn sends the spacecraft on its way to the Moon, leaving the spacecraft with an injection velocity (relative to the synodic frame) of $v_{inj} = 10.92367104$ [km/sec] and located at a phasing angle $\varphi_{inj} = 47.70061087$ [deg] referenced to its LEO orbit as shown below. There is no velocity in the z direction. The coordinate bases of the inertial and synodic frames are aligned at $t = 0$.



NOTE: Use the physical constants and initial conditions exactly as specified, since the problem is extremely sensitive to initial conditions. The spacecraft should make a close approach to the "far side" of the Moon. It is possible - but unlikely if you've done the math correctly - that you'd need to make slight changes to your initial conditions to observe this.

a) Write a function that returns the state derivative as a function of the state, where the position and velocity states are expressed relative to the synodic frame.

HINT: There are at least three ways to get the physical constants (e.g. μ_{Earth}) into your function:

I) You can "hardwire" them into the function code.

II) To make your function reusable for other celestial bodies, you can augment your state vector to include the physical constants as a way of passing them into the function. Those (constant) states should never change, so your function should return a value of zero for their time derivatives. The disadvantages are a larger state vector and a simulation that wastefully computes and stores lots of values that never change.

III) To gain reusability without computing and storing lots of already-known constant values, you can pass the physical constants into the function as extra arguments (not states). This requires use of a MATLAB "anonymous function" when calling **`ode113`**.

Any approach that works is acceptable.

b) Simulate the system in the synodic frame for a period of 6 [days]. For all your simulations, use MATLAB's **`ode113`** numerical integrator with the following options:

```
options = odeset('RelTol', 1e-6, 'AbsTol', 1e-9);
[t_out, y_out] = ode113(@yourfunc, [tstart:tint:tend]', y_initial, options);
```

Plot the orbit in the x, y plane. Include circles on your plot showing the correct locations and radii of Earth and the Moon (see MATLAB's **`viscircles`** command).

c) Do you see anything interesting about the trajectory you just plotted? Why might such a trajectory be useful? To within 0.1 [days], how many days does it take the spacecraft to get from TLI to perilune (i.e. its closest approach to the Moon)?

d) Transform the spacecraft's positions in synodic coordinates into inertial coordinates. Also transform the spacecraft's synodic-frame-relative velocity to inertial velocity expressed in inertial coordinates. Plot the 2D trajectory (derived from the synodic simulation) in the inertial reference frame.

e) Plot effective specific orbital energy $\mathcal{E}_{eff} = -\frac{1}{2}C = -\frac{1}{2}(Jacobi\ constant)$ versus time. Comment on what you see, including any changes in $\mathcal{E}_{eff}$. How large are they, and what might be their cause?

f) Plot specific orbital energy $\mathcal{E}$ versus time. (This requires use of inertial velocity found in part (d).) Comment on what you see, including any changes in $\mathcal{E}$. How large are they, and what might be their cause?

g) Now write a function that returns the state derivative as a function of state, where the position and velocity states are expressed relative to the *inertial* frame. You can assume the inertial and synodic coordinates are identical at $t = 0$.

h) Simulate the system in the *inertial* frame for a period of 6 [days]. To find initial conditions suitable for use in the inertial frame, you'll need to transform the synodic-frame initial conditions. Plot the 2D trajectory (this time derived from the *inertial* simulation) in the inertial reference frame.

i) Compare your trajectories from parts (d) and (h). Comment on any differences you see.

You will reuse some of your code from parts (a) and (b) on Problem Set 2.